Mid-Term - 2026 - 27

Blueprint

Chapter	Teaching Hours	Marks Weightage	Part A 1mark 20 Qs (20 Marks)	Part B 2marks 7 Qs (14 Marks)	Part C 3marks 7Qs (21 Marks)	Part D 5marks 10 Qs (50 Marks)	Total Questions
1. Living World	3	5	3	1	*	*	4
2. Biological Classification	5	9	2	1	*	1	4
3. Plant Kingdom	6	11	1	1	1	1	4
4. Animal Kingdom	9	16	1	1	1	2	5
5. Morphology of Flowering Plants	6	11	1	1	1	1	4
6. Anatomy of Flowering Plants	4	7	2	*	*	1	3
7. Structural Organization in Animals	3	5	2	*	1	*	3
8. Cell: The Unit of Life	11	18	3	1	1	2	7
9. Biomolecules	7	12	2	1	1	1	5
10. Cell Cycle and Cell Division	6	11	3	*	1	1	5
Total	60	105	20	5/7	5/7	5/10	35/44

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MODEL PAPER

General Instructions:

1. The question paper consists of Parts, a, B, C, D and E.
2. Part-A consists of I and II
3. All the parts are compulsory.
4. For Part-A questions, only the answers written first will be considered for evaluation.
5. Draw diagrams wherever necessary. Unlabelled diagrams do not attract any marks.

PART – A

I. Select the correct alternative from the choices given below: $15 \times 1 = 15$

1. Identify the incorrect statement:

   a) Biological names are generally in Latin and written in italics. They are Latinised or derived from Latin irrespective of their origin.

   b) The first word in a biological name represents the genus while the second component denotes the specific epithet.

   c) Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their vernacular origin.

   d) The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter.

2. The process by which organisms are grouped into categories based on easily observable characters is _____

   a) Identification b) Nomenclature c) Classification d) Biodiversity

3. Because of the _____ archaebacteria can endure harsh environments.

   a) They are the oldest living organisms on Earth

   b) They have complicated metabolic processes in their bodies

   c) Rigid cell walls

   d) Double membrane nuclei.

4. The first stage in the development of mosses is

   a) Protonema b) Slender axes c) Gametophyte d) None of the above

5. **Statement I**: Polyp and medusa are the two basic body forms in cnidaria.

   Statement II: Polyps produce medusa sexually and medusa form polyps asexually.

a) Statement I is correct but statement II is incorrect.

b) Statement I is incorrect but statement II is correct.

c) Both statements I and statement II are correct.

d) Both statement I and statement II are incorrect.

6. An example of Perigynous flowers is

a) Brinjal b) Plum c) Guava d) Cucumber

7. Which among the following statements is incorrect about monocotyledonous tissue system?

a) Bulliform cells that help in the curling of leaves are absent in monocot leaf.

b) Epidermis covers the entire plant and protects the plant from external damage.

c) Vascular bundles in monocot stem are neither radial nor conjoint.

d) In a monocot stem pericycle is absent.

8. Identify '3' in the pathway of food ingested by frog.

1. Mouth	→	2. Buccal cavity	→	3. ?	→	4. Oesophagous

a) Larynx b) Pharynx c) Gizzard d) Crop

9. How many pairs of cranial nerves arise from the brain of the frog?

a) 5 b) 7 c) 10 d) 12

10. Live cells were first seen and described by

a) Anton Von Leeuwen hoek b) Robert Hooke

c) Matthias Schleiden d) Theodore Schwann.

11. Amyloplasts are the storage sites of

a) Starch b) Protein c) Oil d) Fat

12. Match the following.

Coloumn 1	*Coloumn 2*
1. Alkaloids	i. Curcumin
2. Essential Oils	ii. Morphine
3. Drugs	iii. Rubber
4. Polymeric substances	iv. Lemon grass oil

a) 1-ii, 2-iv, 3-i, 4-iii b) 1-iii, 2-iv, 3-ii, 4-i

c) 1-ii, 2-i, 3-iii, 4-iv d) 1-i, 2-iii, 3-ii, 4-iv

13. Uridylic acid present in RNA has following components

a) Ribose sugar + Uracil

b) Deoxyribose sugar + Uracil

c) Ribose sugar + Uracil + Phosphate

d) Deoxyribose sugar + Uracil+ Phosphate

14. A stage in cell division is shown in the figure. Select the answer which gives correct identification of the stage with its characteristics.



a) Prophase - Nuclear envelope and Nucleolus disappear.

b) Metaphase – Spindle fibres attach to kinetochores of chromosomes

c) Anaphase – Centromeres split, chromatids separate and move to opposite poles

d) Telophase - Nuclear envelope and Nucleolus reforms.

15. Crossing over takes place between which chromatids and in which stage of the cell cycle?

a) Non sister chromatids of nonhomologous chromosomes at zygotene stage of prophase I.

b) Non sister chromatids of homologous chromosomes at pachytene stage of prophase I.

c) Non sister chromatids of homologous chromosomes at zygotene stage of prophase I.

d) Non sister chromatids of non-homologous chromosomes at pachytene stage of prophase I.

II. Fill in the blanks by choosing the appropriate word from those given in the bracket:

5 x 1 = 5

(bulliform cells, DNA, plasmodesmata, heterocyst, centriole, order)

16. Assemblage of families which exhibit a few similar characters is _____ .

17. *Nostoc* and *Anabaena* can fix atmospheric nitrogen in specialised cells called ____ .

18. Large, empty, colourless cells present in the adaxial epidermis of monocot leaves are called _____ .

19. The cytoplasmic connections between neighbouring plant cells are _____.

20. In animal cells, during the S phase ____ duplicates in the cytoplasm.

PART – B

III. Answer any FIVE of the following questions in 2 to 3 sentences each, wherever applicable:

5 x 2 = 10

21. What is Binomial nomenclature? Give an example.

22. Name the fruiting bodies of a) Ascomycetes b) Basidiomycetes.

23. Differentiate between Rhodophyceae and Phaeophyceae.

24. Mention any two common characters of Annelida and Arthropoda.

25. List the economically important plants of family Solanaceae.

26. Cell wall helps in cell to cell interaction and provides protection. Substantiate.

27. Give an example of a) Homopolysaccharides b) Heteropolysaccharides.

PART – C

IV. Answer any FIVE of the following questions in about 40-80 words each, wherever applicable. **5 × 3 = 15**

28. Identify the following:

    a) Used as fuel and as packing material for trans-shipment of living material.

    b) Used to grow microbes and in preparations of ice creams and jellies.

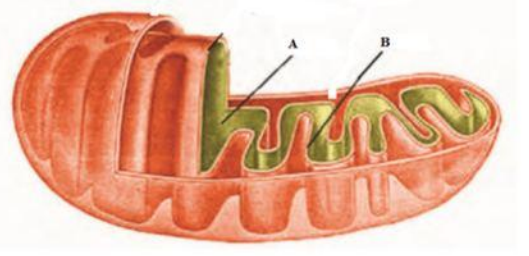
    c) Develops directly from a spore in the life cycle of a moss.

29. Mention any three economically important insects of the phylum Arthropoda, with their economic importance.

30. Draw a labeled diagram showing regions of the root tip.

31. Explain the functions occurring in duodenum of the frog?

32. a) Identify the labelled parts A, B in the given diagram. 2



    b) Why mitochondria are regarded as 'power houses' of the cell? 1

33. Lipids are not strictly macromolecules. Substantiate.

34. Enumerate the significance of Mitosis.

PART – D

V. Answer any FOUR of the following questions in 200–250 words each, wherever applicable. **4 × 5 = 20**

35. Explain the salient features of kingdom Monera

36. "Algae are useful to man in a variety of ways". Justify the statement with suitable examples.

37. Give the identifying features of the phylum Echinodermata.

38. Draw a neat diagram of transverse section of dicot root and label any five parts.

39. Write any five differences between prokaryotic cell and eukaryotic cell.

40. List any five functions of proteins with examples.

41. With respect to meiosis define:

a) Bivalent b) Recombination nodules c) Crossing over

d) Recombination e) Chiasmata

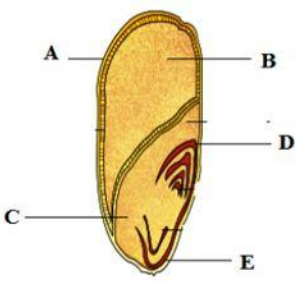
VI. Answer any ONE of the following questions in 200–250 words each, wherever applicable.

$1 \times 5 = 5$

42. Given below is the table representing name of the phylum, organism and the unique organ belongs the phylum. Complete the boxes in the table labelled as a, b, c, d and e with appropriate answer.

Arthropoda	Cockroach	a
b	Pleurobrachia	Comb plate
Mollusca	Apple snail	c
Porifera	d	Spongocoel
e	Neries	Parapodia

43. Answer the following questions with respect to diagram given below:



a) Identify the outermost layer of the seed marked A 1

b) Name the part B that stores food. 1

c) Name the single cotyledon C 1

d) Name the sheaths D & E that enclose the plumule and radicle respectively. 2

44. The micro tubules in flagella of a eukaryotic cell are arranged in a particular fashion. Illustrate with a diagram.

PART – E

(For visually challenged students only)

14. In meiosis crossing over is initiated at

a) Zygotene b) Diplotene c) Pachytene d) Leptotene

32. Cell wall is non living rigid structure made of three layers. Mention the wall layers and its chemical composition.

43. a) What is the pattern of arrangement of leaves on stem called? 1

b) Mention the three types of arrangement of leaves. 3

c) Mention the type of arrangement of leaf observed in *Calotropis*. 1

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